



Understanding the effects of multiphase flow properties in transport-limited organic electrosynthesis

Casey K. Bloomquist^a, Ziqi Zhang^a, Eray S. Aydil^a, Miguel A. Modestino^{a,*}

^a New York University, Tandon School of Engineering, Department of Chemical and Biomolecular Engineering, 6 Metrotech Center, Brooklyn, NY, 11201, USA

ARTICLE INFO

Keywords:

Multiphase flow
Electrosynthesis
Adiponitrile
Mass transport
Bubble-induced convection

ABSTRACT

Achieving industrial viability of organic electrosynthesis requires overcoming mass transport limitations that compromise both selectivity and production rates. While multiphase operation above the solubility limit provides opportunities to enhance mass transport, the relationship between operating conditions, flow regimes, and electrochemical performance remains poorly understood. Here, we systematically investigate multiphase electrochemical systems using acrylonitrile electrohydrodimerization as a model reaction. Through flow visualization and data-driven surrogate modeling, we explore the effects of flow regime, interelectrode gap, organic fraction, current density, and flow rate on performance. Our results reveal that bubble-induced convection dominates over flow regime effects, enabling maintenance of high faradaic efficiency (73–76 %) at elevated current densities ($> 200 \text{ mA cm}^{-2}$) and achieving production rates up to $5 \times 10^{-4} \text{ kg cm}^{-2} \text{ h}^{-1}$ without selectivity penalties. This contrasts with conventional systems where increased current density reduces selectivity. Multi-objective optimization reveals key design guidelines: moderate interelectrode gaps balance mixing with ohmic losses, operation above the solubility limit enhances transport without requiring maximum concentrations, and lower flow velocities prove advantageous. The methodology developed combines flow visualization, dimensionless analysis, and data-driven modeling to provide a framework for studying other electrochemical reactions with similar mass transport limitations, contributing to the development of more efficient organic electrochemical processes for sustainable chemical manufacturing.

1. Introduction

Electrochemical processes offer a more sustainable approach to chemical manufacturing by directly using electricity instead of heat from fossil fuels [1–4]. As the global chemical industry increasingly seeks sustainable alternatives to conventional production methods, organic electrochemistry represents a viable pathway. Approximately 75 % of industrial chemicals are organic compounds, yet their production relies almost exclusively on thermochemical routes [5]. However, organic electrochemical processes face inherent challenges in achieving the selectivity and production rates required for industrial applications due to mass transport limitations [6–11]. The limited solubility of organic reactants in aqueous electrolyte solutions constrains reaction rates and can negatively impact product selectivity. Flow electrosynthesis provides an effective approach to address these limitations by providing precise control over residence time, enhanced mass transfer, and the ability to operate under steady-state conditions [5,11,12]. Previous work from our group demonstrated that convection can partially

overcome these limitations by enhancing mass transport, with both forced liquid convection and bubble-induced convection capable of enhancing reaction performance in similar ways by producing equivalent mass transport conditions [13]. Flow microreactors further amplify these benefits by decreasing interelectrode gaps to submillimeter to minimize ohmic resistance while maintaining high surface-to-volume ratios for improved mass transport [12,14].

Building on these flow-based approaches, multiphase systems offer another avenue for mass transport enhancement by enabling operation above the solubility limit of organic reactants [15]. The structure of multiphase flow, characterized by flow regimes such as droplet, slug, and parallel flows, influences the interfacial area between phases and the degree of mixing, all of which affect reactant transport to the electrode surface and thus the selectivity between desired products and competing side reactions. However, multiphase systems introduce additional complexity while engaging several distinct mass transport mechanisms simultaneously [16]. These mechanisms, shown in Fig. 1 A–D include organic phase partitioning that provides a continuous

* Corresponding author.

E-mail address: modestino@nyu.edu (M.A. Modestino).

<https://doi.org/10.1016/j.cej.2025.172125>

Received 2 September 2025; Received in revised form 18 November 2025; Accepted 19 December 2025

Available online 20 December 2025

1385-8947/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

reservoir of reactants, organic phase disturbance that promotes mixing, forced convection from bulk flow, and bubble-induced convection from gas evolution at the electrodes. Multiphase flows also negatively impact performance: gas bubbles and organic droplets can block conduction pathways and increase ohmic resistance, cover electrode surfaces, span the interelectrode gap, and potentially alter product distributions. While multiphase flow has been extensively studied in non-electrochemical systems, with well-established correlations for mass transfer coefficients and flow regime transitions [17–26], the relationship between multiphase flow structure and electrochemical performance remains poorly understood, particularly for organic synthesis reactions under realistic operating conditions. Key questions remain: Can controlled two-phase flows be implemented to enhance mass transport rates in electrochemical reactions involving organic reactants with limited solubility in electrolytes? What is the relationship between multiphase flow structure and both the diffusive transport rates of reactants to the electrodes and ohmic losses? Furthermore, is multiphase flow structure the defining characteristic of reactor performance, or do other operational parameters—such as current density, organic fraction, interelectrode gap, and flow velocity—play equally important or dominant roles in determining electrochemical outcomes?

In this study, we aim to provide insights into these questions by systematically investigating multiphase electrochemical systems using the industrially relevant electrohydrodimerization of acrylonitrile (AN) to adiponitrile (ADN) as a model reaction. The electrochemical synthesis of ADN represents one of the largest organic electrochemical processes in industry, serving as a key precursor for nylon manufacturing [2,27–29]. The desired cathodic reaction involves the reductive dimerization of two AN molecules to form ADN, but this process competes with several undesired pathways (Fig. 1E). At the cathode, AN undergoes electron transfer to form radical anion intermediates that can either dimerize to produce the desired ADN product or follow alternative reaction pathways leading to side products. Competing reactions include further reduction to propionitrile (PN), oligomerization to form trimers and higher oligomers, and hydrogen evolution. Mass transport conditions and the balance between diffusion and electrochemical reaction rates strongly influence the selectivity toward ADN versus these side products [30]. We quantify this selectivity using the Faradaic efficiency toward ADN (FE_{ADN}), which represents the fraction of total charge that produces ADN. Similarly, we quantify side product

formation through Faradaic efficiencies toward PN (FE_{PN}) and tricyanohexane (TCH, FE_{TCH}). At low current densities with high AN surface concentrations, oligomer formation is favored, while at high current densities where AN becomes depleted at the electrode surface, PN formation and hydrogen evolution are promoted. The desired ADN formation occurs under intermediate conditions where there is an optimal balance between mass transport and reaction kinetics.

To understand how multiphase flow affects electrochemical outcomes, we use flow visualization to characterize flow regimes and construct flow regime maps that connect hydrodynamic conditions to electrochemical performance. We then systematically explore how different flow regimes influence ADN Faradaic efficiency (FE_{ADN} , the selectivity toward the desired product) under various operating conditions. Our findings show that while flow regime has a minor impact on performance, the major contributors are bubble-induced convection, interelectrode spacing, and acrylonitrile volume fraction above the solubility limit. Through Pareto analysis, we identify optimal operating windows that balance production rate and energy productivity, providing quantitative design guidelines for multiphase electrochemical reactors.

2. Characterization of multiphase flow regimes

Our experimental setup (Fig. 2) used a parallel-plate electrochemical flow reactor with a cadmium foil cathode and stainless steel anode. A serpentine flow path separated the electrodes, with interelectrode gaps of 0.25 to 1.0 mm controlled by interchangeable gaskets to explore the range typical of microreactor designs. Spring test probes contacted the electrode backs through end plates, enabling current densities ranging from 70 to 300 mA cm⁻². Independent syringe pumps delivered the aqueous and organic phases to a T-junction before the reactor, allowing precise flow rate control and enabling exploration of different flow regimes. We investigated total flow rates from 2 to 10 mL min⁻¹ to access a range of flow velocities within the operational limits of our pumps and reactor design. The aqueous phase contained 0.5 M sodium phosphate buffer, 0.03 M ethylenediaminetetraacetic acid (EDTA) as a chelating agent, and 0.02 M tetrabutylammonium hydroxide (TBA) for selectivity enhancement. The organic phase was pure acrylonitrile (AN), with organic volume fractions ranging from 0.05 to 0.3 to cover conditions from below the AN solubility limit (0.09) to well above, with the upper

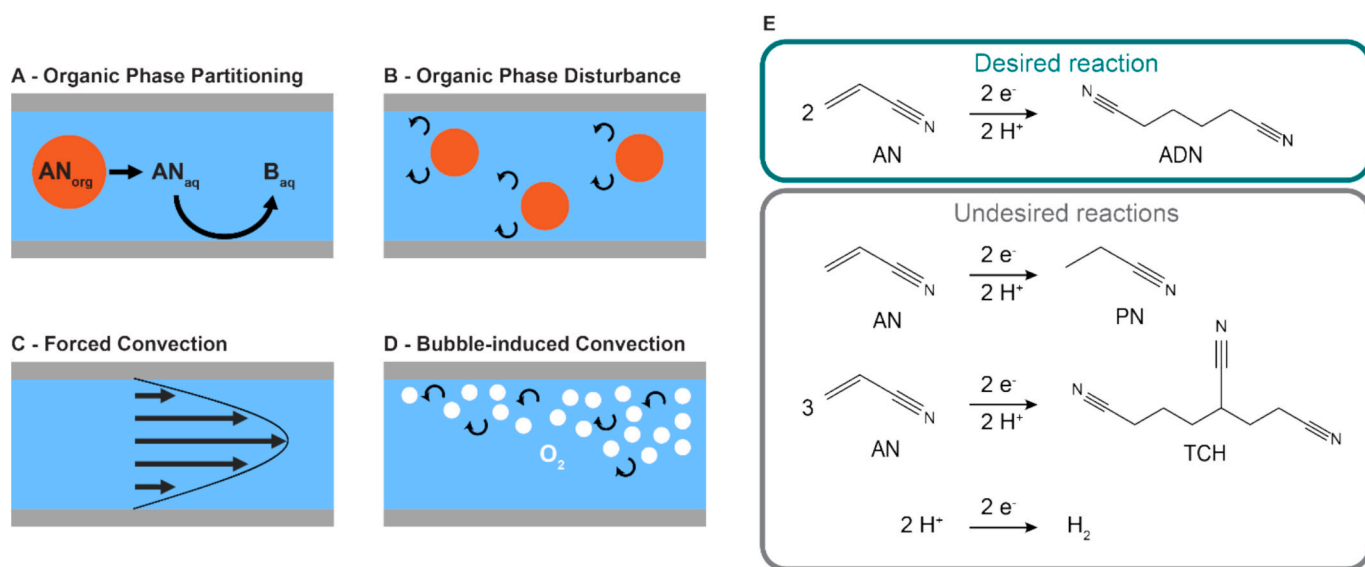


Fig. 1. (A–D) Contributions to mass transport in multiphase electrochemical systems including (A) organic phase partitioning into the aqueous phase, (B) organic phase disturbance, (C) forced convection, and (D) bubble induced convection from oxygen evolved at the anode or hydrogen evolved at the cathode. (E) Desired and undesired cathodic reactions in the electrohydrodimerization of acrylonitrile (AN) to adiponitrile (ADN). Competing reactions include the production of propionitrile (PN), 1,3,6 tricyanohexane (TCH), and the hydrogen evolution reaction (HER).

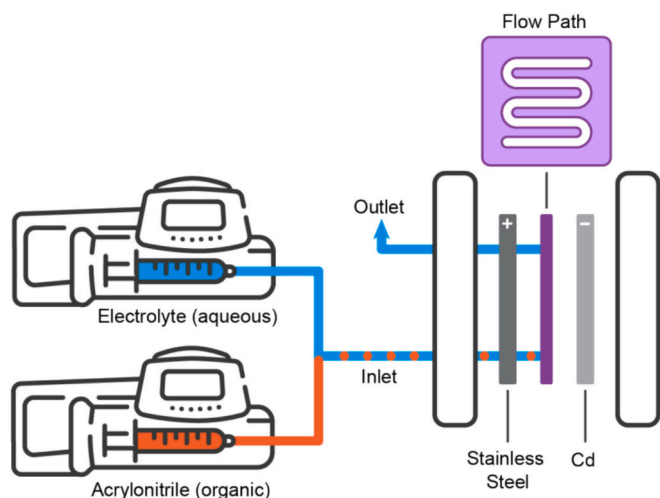


Fig. 2. Experimental setup. The flow reactor (Analytical Sales) consists of a serpentine flow path sandwiched between a stainless steel (anode) and cadmium (cathode) electrode. Electrical contact is made with a spring test probe through the back of the end plates (not shown). Streams of pure electrolyte (aqueous) and pure acrylonitrile (organic) are pumped by syringe pumps and combined in a T-fitting and sent into the reactor. The mixed stream flows through the flow path in a single pass and is collected at the outlet for extraction and analysis.

limit set to avoid excessive solid formation that could clog the reactor or cause electrical shorting. The mixed stream passed through the reactor once and was collected for extraction and analysis.

To observe the multiphase flow regimes, we replaced the metal end

plate with a clear acrylic plate and the metal electrode with a glass plate. The glass plate was plasma-treated to increase its hydrophilicity and closely resemble a cadmium electrode. We added fluorescein dye to the aqueous phase and observed flow under UV illumination. Using a high-speed camera, we captured videos and images of the flow across the complete parameter space (AN Fraction: 0.05, 0.1, 0.15, 0.2, 0.25, 0.3; total flow rate: 2, 4, 6, 8, 10 mL min⁻¹; interelectrode gap: 0.25, 0.5, 1.0 mm) and visually characterized the flow regime for each condition. Representative images of the flow regimes are shown in Fig. 3. We identified four flow regimes: droplet flow, slug flow, parallel flow, and transition flow. Droplet flow consisted of spherical organic droplets (0.5–2 mm diameter) with minimal wall contact. Slug flow presented elongated organic structures (> 2 mm length) that spanned the interelectrode gap. Parallel flow featured both phases flowing side-by-side as continuous streams with a defined interface. Transition flow exhibited characteristics of multiple regimes, representing dynamic evolution between flow patterns. These flows often showed temporal instabilities, fluctuating between patterns based on local conditions.

We used the Weber number—a dimensionless parameter that characterizes the balance between inertial forces and surface tension forces—to map the flow regimes. The Weber number for each phase is defined as:

$$We_i = \frac{\rho_i v_i^2 l}{\sigma}$$

where ρ_i is the density of phase i , v_i is the superficial velocity of phase i , l is the interelectrode gap, and σ is the interfacial tension between the two phases (approximately $\sigma \approx 10.5$ mN m⁻¹, estimated using the Girifalco-Good correlation as detailed in the Supplementary Information). Weber numbers were calculated using superficial velocities from individual phase flow rates and reactor cross-sectional area, which depends on the

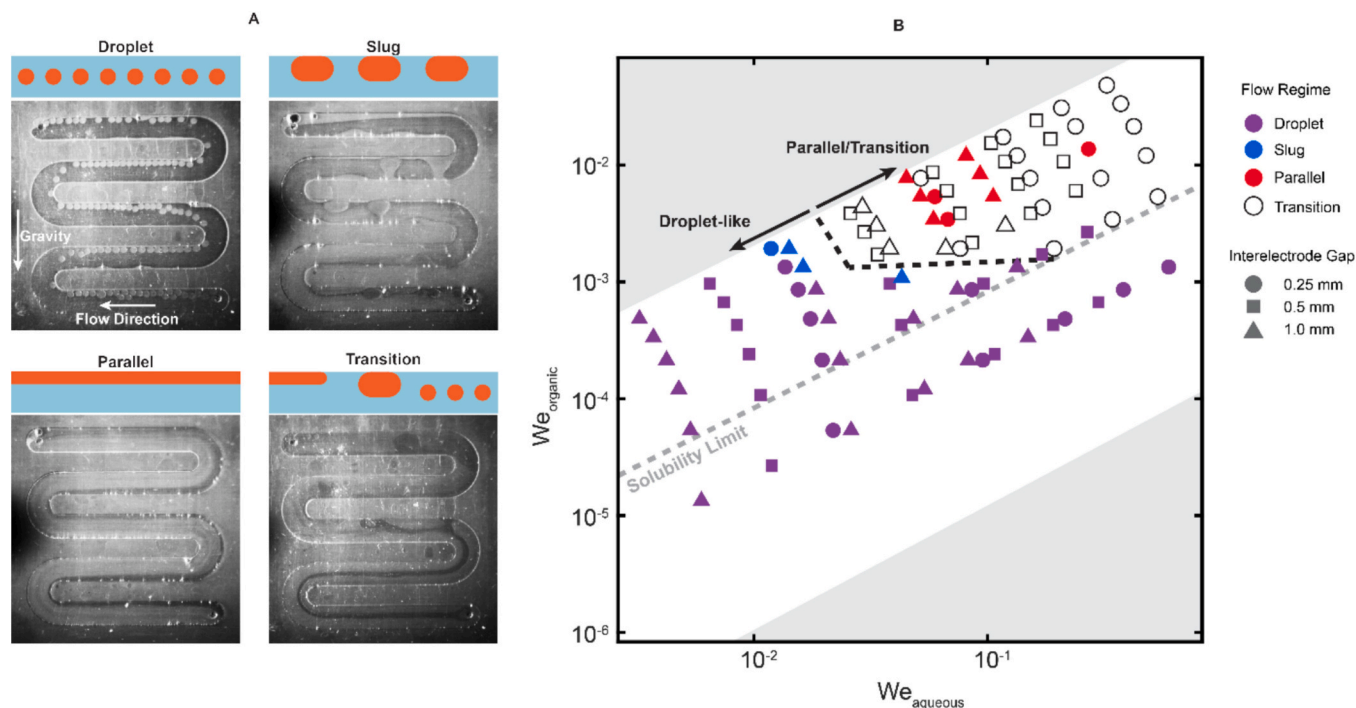


Fig. 3. (A) Visualizing flow regime inside the reactor using fluorescent dye and UV light. Observed flow regimes including droplet-like flows (droplet flow, slug flow), parallel-like flows and transition flows. (B) Flow regime map showing observed flow regime as a function of aqueous and organic Weber number (We). Observations were made across the range of AN volume fraction (0.05–0.3), total flow rate (2–10 mL min⁻¹) and interelectrode gap (0.25–1.0 mm). The flow rates (aqueous and organic) and interelectrode gap were used to calculate We_{aq} and We_{org} . The videos at each condition were visually characterized as droplet, slug, parallel, or transition flow, and are indicated by the symbol color. To distinguish the various interelectrode gaps, the different gaps are represented by different shapes (● – 0.25 mm, ■ – 0.5 mm, ▲ – 1.0 mm). The AN solubility limit is shown as a gray dashed line. The black dashed line represents the boundary between the droplet-like flow regime and the parallel/transition flow regime. The flow regime map is constructed from flow visualization experiments without active electrochemical reaction.

interelectrode gap. We constructed a flow regime map plotting We_{aqueous} versus We_{organic} for the various experimental conditions, shown in Fig. 3B, where the data point color represents the observed flow regime and the data point shapes represent the interelectrode gap (● – 0.25 mm, ■ – 0.5 mm, ▲ – 1.0 mm). This approach enabled us to classify flow regimes across different operating conditions and reactor geometries. Droplet flow is the primary flow regime for all interelectrode gaps below the solubility limit (dashed gray line in Fig. 3B). Above the solubility limit, a boundary (dashed black line) separated droplet-like flows (droplet and slug flow) from parallel and transition flows. Droplet-like flows occurred at lower organic Weber numbers ($We_{\text{organic}} < \sim 10^{-3}$), corresponding to lower superficial velocities, while parallel and transition flows occurred at higher organic and aqueous Weber numbers ($We_{\text{organic}} > \sim 10^{-3}$, $We_{\text{aqueous}} > \sim 10^{-1.5}$) and higher superficial velocities. By mapping performance metrics such as Faradaic efficiency and production rate as functions of Weber numbers, we can identify how different flow regimes influence mass transport to the electrode surface and thus reaction selectivity.

While the flow regime map provides a framework for baseline flow patterns, electrochemical operation introduces significant complexity. Gas bubble generation from electrolyte reactions (hydrogen evolution at the cathode, oxygen evolution at the anode) dramatically alters reactor hydrodynamics. Bubble-induced convection breaks up the organic phase and enhances mixing between phases. The added gas volume increases ohmic resistance by blocking conductivity pathways, alters local current density by covering electrode surfaces, and shifts the flow regime by increasing superficial fluid velocities. Additionally, the serpentine flow path creates local variations in flow structure that cannot be fully characterized during active electrochemical operation. Despite these complications, our flow regime map serves as a baseline for understanding initial multiphase structure and correlating that initial structure with performance trends.

3. Data-driven systematic exploration of multiphase flow effects

We used a data-driven approach with surrogate modeling to investigate multiphase flow effects on ADN electrosynthesis. Our experimental design (Fig. 4) followed a multi-step process to explore the parameter space efficiently. We used Hammersley sampling, a quasi-random low-discrepancy sampling method, to select experimental conditions spanning current densities from 70 to 300 mA cm⁻² and AN fractions from 0.05 to 0.3 (covering conditions from below the AN solubility limit of 0.09 to well above). Hammersley sampling generates

uniformly distributed points across the parameter space with more even spacing than random sampling, ensuring comprehensive coverage of the operational range of current density and AN fraction. We performed reactions at these conditions for nine reactor configurations, representing every combination of interelectrode gap (0.25, 0.5, 1.0 mm) and total flow rate (2, 6, 10 mL min⁻¹), yielding 162 total experimental conditions (18 Hammersley-sampled combinations of current density and AN fraction $\times$ 9 reactor configurations). We calculated Weber numbers for both phases using the superficial velocities, fluid properties, and reactor dimensions for each of the experimental conditions. This transformed our data into a dimensionless framework for comparison with the flow regime map with We_{aqueous} ranging from $\sim 10^{-3}$ to ~ 1 and We_{organic} ranging from $\sim 10^{-5}$ to $\sim 10^{-2}$.

We constructed Gaussian process regression (GPR) surrogate models to predict performance metrics across the entire three-dimensional parameter space defined by We_{aqueous} , We_{organic} , and current density. For each performance metric (FE_{ADN} , FE_{TCH} , FE_{PN} , ADN production rate, and ADN energy productivity) and each interelectrode gap, we trained separate GPR models using the experimental measurements as training data. The GPR algorithm takes as inputs the operational parameters ($X = [We_{\text{aqueous}}, We_{\text{organic}}, \text{current density}]$) and the corresponding measured performance metric ($Y = FE_{\text{ADN}}$, for example), then constructs a non-parametric regression model that predicts performance at any point in the parameter space. The trained models generate continuous predictions by interpolating between the discrete experimental measurements using the Gaussian process covariance structure, which enables smooth predictions across the parameter space, enabling systematic analysis of how Weber numbers and current density jointly influence performance. For more details on the GPR implementation, including kernel selection and hyperparameter tuning, refer to the Supplementary Information (Section “Building Surrogate Models”). Model validation was performed by comparing measured experimental values with GPR predictions. The models demonstrate strong predictive accuracy, with coefficients of determination (R^2) ranging above 0.99 and root mean square errors (RMSE) of 2.20 to 2.78 for FE_{ADN} across the three interelectrode gaps (see Fig. S3-S5 in Supplementary Information for comprehensive validation plots). Complete experimental conditions and measured values for all 162 experiments are provided in Tables S1-S10 of the Supplementary Information to ensure reproducibility.

We plotted results in three-dimensional space using organic Weber number, aqueous Weber number, and current density as axes. We created mathematical regression models using Gaussian process regression (GPR) to predict performance across the entire 3D parameter

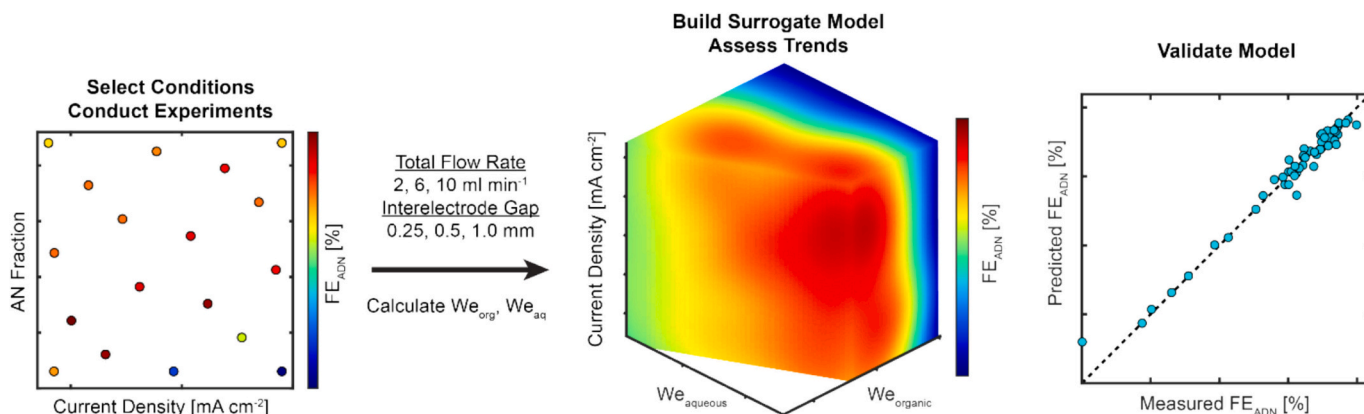


Fig. 4. Process for developing data-driven surrogate models. Hammersley sampling selects 18 pseudorandom experimental conditions for current density (70–300 mA cm⁻²) and AN fraction (0.05–0.3). Electrochemical experiments are performed at these conditions across all 9 combinations of total flow rate (2, 6, 10 mL min⁻¹) and interelectrode gap (0.25, 0.5, 1.0 mm), yielding 162 total experimental conditions. Organic and aqueous Weber numbers (We_{organic} and We_{aqueous}) are calculated from the flow conditions, enabling Gaussian process regression (GPR) to build surrogate models that predict performance metrics (FE, production rate, energy productivity) as continuous functions of We_{aqueous} , We_{organic} , and current density for trend analysis. Model validation involves assessing prediction standard deviations across the experimental space and comparing measured versus predicted values using R^2 and RMSE metrics.

space. GPR provides uncertainty quantification through prediction variance, validating model reliability, and potentially identifying regions needing additional experiments. The models can be further validated by comparing measured and predicted FE values (see Supplementary Information Fig. S3-S5). The trained models generated continuous predictions across the parameter space, enabling systematic analysis of how Weber numbers and current density jointly influence performance.

3.1. Impact of flow regimes on electrochemical performance

Having established our 3D surrogate models, we can now examine how multiphase flow regimes and operational parameters such as current density, interelectrode gap, organic fraction, and superficial velocity affect electrochemical performance. We address two key questions: first, whether the flow regime boundaries identified in our Weber number maps translate into meaningful performance differences, and second, how other factors—including mass transport effects, bubble formation, and operational parameters—influence electrochemical outcomes. Fig. 5 illustrates the ADN Faradaic efficiency (FE_{ADN}) as a function of $We_{aqueous}$, $We_{organic}$, and current density. The top row shows 3D visualizations for each interelectrode gap, while the bottom panels show 2D slices at three current densities (90, 190, and 290 mA cm⁻²) that reveal how performance varies across the flow regime map at fixed electrochemical driving forces. Similar plots for FE_{TCH} and FE_{PN} are included in the Supplementary Information (Fig. S6 and S7).

Flow regime effects are visible in the 2D slices presented in Fig. 5. At low current density (90 mA cm⁻²), the 0.25 mm gap shows the highest FE_{ADN} (65 %) in parallel/transition flow, while 0.5 and 1.0 mm gaps exhibit high FE_{ADN} (65–75 %) in both droplet-like and parallel/transition regimes when organic loading approaches the solubility limit. As current density increases to 290 mA cm⁻², the high FE_{ADN} regions shift toward higher organic loading (higher $We_{organic}$) as more AN is required to balance reaction and mass transport rates. The high FE_{ADN} regions also shift toward droplet-like flow regimes. However, maximum FE_{ADN} in parallel/transition flow is only 10–15 % lower than in droplet-like flow, though it decreases more significantly as Weber numbers increase.

An interesting finding is that high FE_{ADN} (73–76 %) is maintained for all interelectrode gaps as current density increases from 90 to 290 mA cm⁻². This contrasts with conventional single-phase adiponitrile electrosynthesis, where FE_{ADN} typically decreases with increasing current density due to increased mass transport limitations and limited organic concentration [13,30]. The high FE_{ADN} at high current densities suggests that bubble-induced convection plays a critical role in determining performance in multiphase electrochemical systems. Gas bubbles generated from electrolyte reactions create turbulent mixing that effectively disrupts the organic phase and enhances mass transport of organic reactants to the electrode surface. This creates a beneficial synergy where increased current density generates more bubbles, which in turn provide the enhanced mixing needed to sustain high selectivity at those same elevated current densities.

Even though flows exhibit similar regimes across different interelectrode gaps, both 3D visualizations and 2D slices (Fig. 5) reveal that each gap shows distinct performance patterns that depend on factors beyond flow regime alone. While all interelectrode gaps achieved similar maximum FE_{ADN} values, larger gaps (0.5 and 1.0 mm) resulted in a broader range of Weber number combinations that can achieve FE_{ADN} (> 60 %) than the 0.25 mm gap. At high current densities where mass transport limitations are most severe, the 0.25 mm gap requires maximum organic fraction to achieve high performance. Smaller interelectrode gaps create several performance limitations. The reduced reaction volume causes organic droplets and gas bubbles to span the entire gap and occupy a large fraction of the reactor volume. This creates locally elevated current densities that drive undesired side reactions, increases effective superficial velocities that push flow toward parallel regimes, and increases ohmic losses that reduce energy efficiency. The

benefits of bubble-induced convection are also strongly dependent on the interelectrode gap. While larger gaps (0.5 and 1.0 mm) benefit from bubble-generated mixing that enhances mass transport, the smallest gap (0.25 mm) shows significantly diminished benefits. In smaller reactor volumes, the negative consequences of gas generation, including large gas void fractions that block conduction pathways and create locally elevated current densities, overcome the mass transport improvements. This demonstrates that reactor design must balance the beneficial mixing effects of bubbles against their detrimental impact on current distribution and ohmic resistance.

Operating at or above the solubility limit proves beneficial for achieving high FE_{ADN} across all conditions. However, the optimal organic fraction depends strongly on both current density and interelectrode gap, reflecting the interplay between mass transport processes and the rate of electrochemical reactions at the electrode surface. At lower current densities (90 mA cm⁻²), where mass transport limitations are lower, larger gaps (0.5 and 1.0 mm) achieve high FE_{ADN} at the solubility limit. In comparison, the 0.25 mm gap requires higher organic fractions even at low current densities. As current density increases to 290 mA cm⁻², all interelectrode gaps require higher organic fractions to maintain high selectivity, with maximum FE_{ADN} regions shifting toward higher $We_{organic}$ values. This shift reflects the need for increased organic reactant availability to balance elevated electrochemical reaction rates. These findings suggest that the organic fraction is more important than the flow regime for achieving high performance. The critical requirement is having sufficient AN available at the electrode surface rather than optimizing the incoming flow structure. This is particularly true at elevated current densities where bubble-induced convection provides significant mixing regardless of the initial flow pattern.

When comparing the relative importance of these parameters, several key insights emerge. Flow regime has a measurable effect on performance, with droplet-like flows generally preferred and maximum FE_{ADN} in parallel/transition flow only 10–15 % lower than in droplet-like flow, however the magnitude of this effect is small compared to the influence of current density, interelectrode gap, and organic fraction. The critical requirement is having sufficient AN available rather than optimizing the incoming flow structure, particularly at elevated current densities where bubble-induced convection provides significant mixing regardless of the initial flow pattern. While operational parameters such as interelectrode gap, flow rate, organic fraction, and current density all influence flow regime, they also affect electrochemical performance through mechanisms beyond flow regime classification alone. For instance, interelectrode gap affects ohmic losses and bubble confinement, current density influences bubble generation and electrode overpotentials, and substantial performance variation exists within single flow regime. Our analysis treats these parameters and their coupled effects holistically rather than assuming flow regime is the primary determinant of performance—an approach supported by the findings above that bubble-induced convection and other factors often dominate over flow regime distinctions.

4. Understanding the factors that lead to optimal multiphase electrochemical performance

While FE_{ADN} trends establish the fundamental performance landscape, industrial viability requires examining production rate and energy productivity across the same parameter space. Production rate, measured as the mass of ADN produced per unit electrode area per time, determines process throughput and capital equipment utilization. Energy productivity, measured as the mass of ADN produced per unit energy, directly affects operating costs and environmental footprint. These two metrics exhibit competing trends: production rate generally increases with current density due to higher electrochemical reaction rates, while energy productivity typically decreases with current density due to increased ohmic losses and concentration overpotentials. Understanding these fundamental tradeoffs and identifying operating

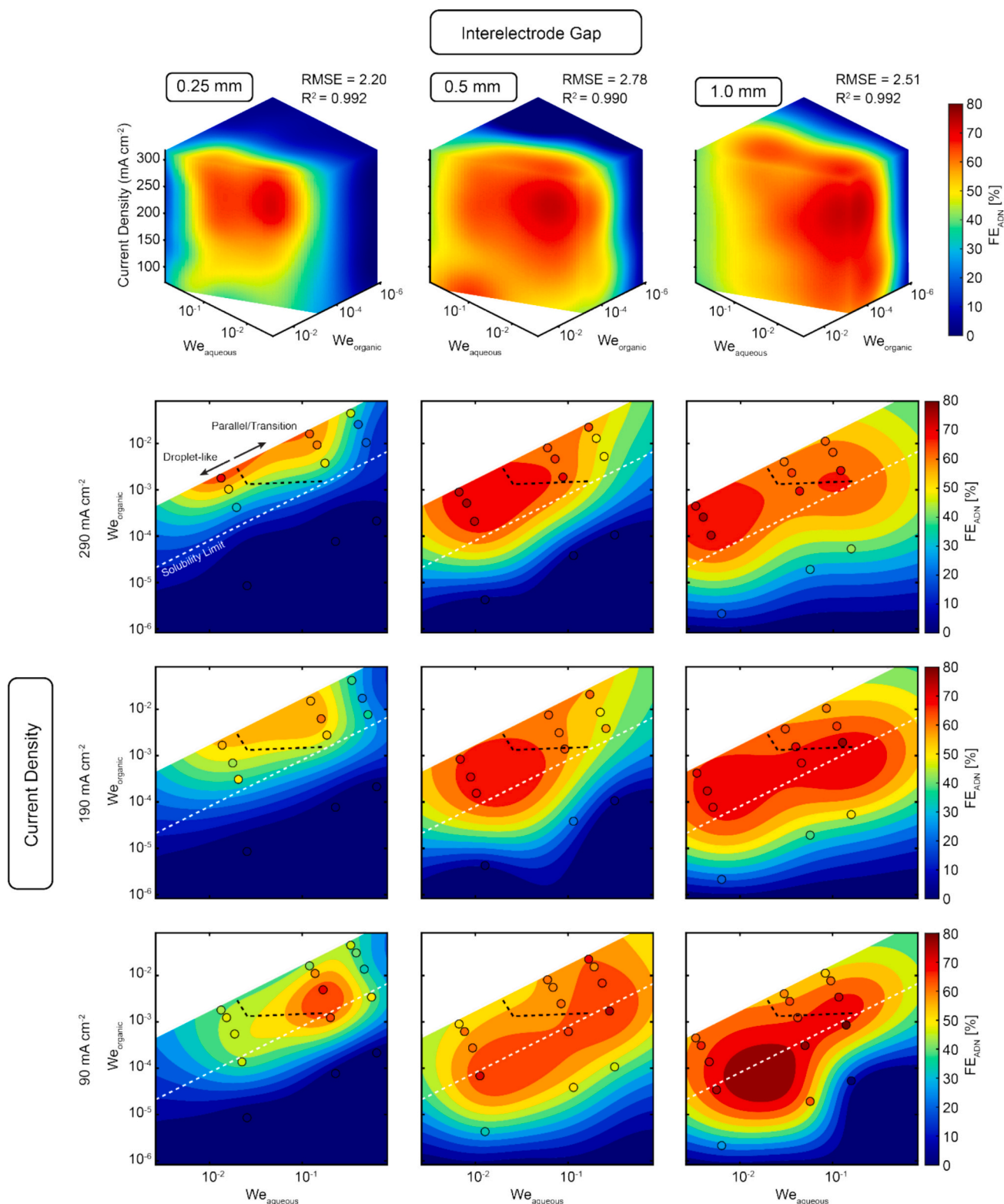


Fig. 5. ADN Faradaic efficiency (FE_{ADN}) as a function of aqueous Weber number, organic Weber number, and current density for different interelectrode gaps. The color scale represents FE_{ADN} values, with circular markers in 2D slices indicating measured experimental data points and the colored background showing Gaussian Process Regression model predictions. The top row displays 3D visualizations for each interelectrode gap (0.25, 0.5, and 1.0 mm), while the bottom panels show 2D slices at three current densities (90, 190, and 290 mA cm^{-2}) revealing performance variations across flow regime boundaries. The data points in the 2D slices represent experimental measurements from current densities within $\pm 30 \text{ mA cm}^{-2}$ of slice value. High efficiency regions ($> 70\%$) occur primarily in droplet-like flow regimes at low We_{aqueous} and high We_{organic} conditions. Maximum Faradaic efficiency remains stable across all current densities, with optimal regions shifting toward higher organic loading at elevated current densities. The black dashed line indicates the transition from droplet-like to parallel-like flow regimes and the white dashed line indicates the AN solubility limit. Flow regime effects are measurable but modest compared to the influence of current density, interelectrode gap, and organic fraction.

conditions that optimally balance competing objectives is essential for industrial implementation.

Pareto analysis revealed optimal operating conditions and quantified key performance tradeoffs. Pareto analysis is a multi-objective optimization technique that identifies the set of solutions where improvement in one objective cannot be achieved without degrading another objective. In the context of electrochemical processes, this approach is particularly valuable for understanding the inherent tradeoffs between production rate and energy productivity. The Pareto front represents the boundary of optimal achievable combinations, guiding industrial process design and operation. Fig. 6 presents our Pareto analysis results, showing the optimal tradeoffs between production rate and energy productivity across all experimental conditions. Fig. 6A displays the overall Pareto front, where gray points represent all experimental data points and the red points and line represent the optimal achievable conditions. The green square represents the maximum production rate condition ($4.28 \times 10^{-4} \text{ kg cm}^{-2} \text{ h}^{-1}$) while the pink triangle represents the maximum energy productivity condition ($0.486 \text{ kg kWh}^{-1}$). The panels (Fig. 6B-E) show the same data but examine how different operating parameters (current density, interelectrode spacing, organic fraction, and superficial velocity) affect the Pareto front.

Current density has a major influence on the Pareto front, as shown in Fig. 6B, where optimal solutions are separated by current density levels. Maximum energy productivity occurs at lower current densities ($< 140 \text{ mA cm}^{-2}$), while maximum production rates require high

current densities ($> 230 \text{ mA cm}^{-2}$). When we increase the current density and move from maximum energy productivity to maximum production rate conditions, we observe a 182 % increase in production rate (from 1.52×10^{-4} to $4.28 \times 10^{-4} \text{ kg cm}^{-2} \text{ h}^{-1}$) coupled with a 30 % decrease in energy productivity (from 0.486 to 0.34 kg kWh^{-1}). This substantial improvement in throughput comes at a moderate energy penalty, suggesting favorable economics for many industrial applications. The relationship between energy productivity and current density reflects competing effects. While higher current densities increase ohmic losses and concentration overpotentials, they also influence reaction selectivity by altering the relative rates of ADN formation versus competing reactions (PN, TCH, and HER formation). These selectivity effects are captured in the production rate (Eq. S5), which is the product of Faradaic efficiency toward ADN (FE_{ADN}) and current density, such that shifts in selectivity due to overpotential changes directly manifest as changes in production rate. At low current densities, energy productivity decreases rapidly with increasing current density due to ohmic losses, concentration overpotentials, and changing reaction selectivity. However, above $\sim 140 \text{ mA cm}^{-2}$, production rate increases substantially while energy productivity decreases only slightly, likely because enhanced bubble-induced convection maintains high Faradaic efficiency toward ADN without dramatically affecting energy efficiency. This creates a fundamental design choice: operators can prioritize energy efficiency by operating at low current densities or maximize throughput by operating at high current densities, with particularly favorable

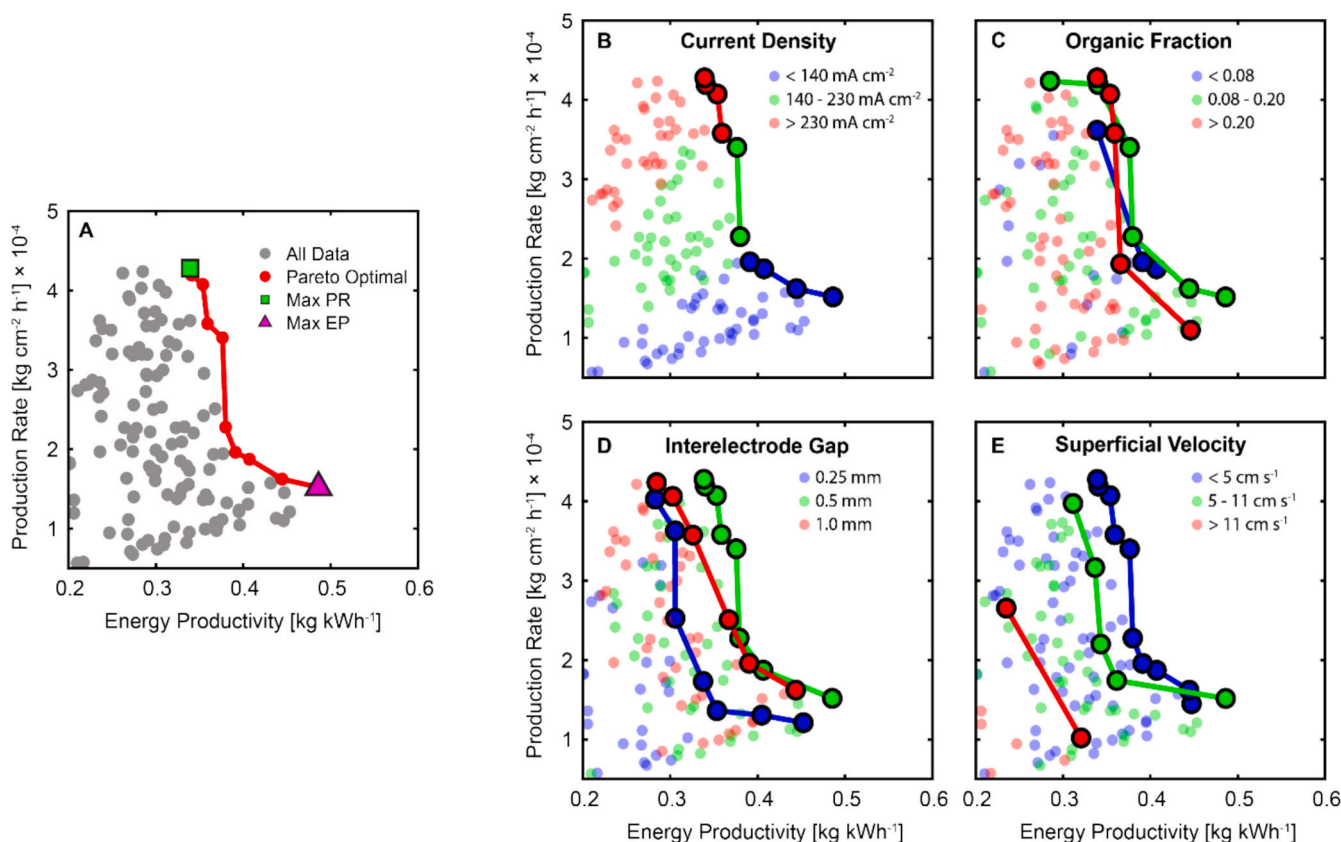


Fig. 6. Multi-objective Pareto analysis revealing optimal tradeoffs between ADN production rate and energy productivity across all experimental conditions. Panel A displays the overall Pareto front, where gray points represent all experimental data and red points define the optimal achievable combinations. The green square indicates maximum production rate conditions ($4.28 \times 10^{-4} \text{ kg cm}^{-2} \text{ h}^{-1}$) while the pink triangle shows maximum energy productivity conditions ($0.486 \text{ kg kWh}^{-1}$). Moving from maximum energy productivity to maximum production rate yields a 182 % increase in throughput with only a 30 % decrease in energy efficiency. Panels B-E examine how different operating parameters affect the Pareto front: (B) current density strongly influences optimal solutions, with maximum energy productivity occurring below 140 mA cm^{-2} and maximum production rates requiring above 230 mA cm^{-2} ; (C) organic fraction shows similar performance across all ranges, with operation above solubility limits essential for highest production rates; (D) interelectrode gap significantly impacts performance, with 0.5 mm providing optimal balance across the entire front while 0.25 mm shows poor performance despite lower ohmic resistance; (E) superficial velocity reveals counterintuitive preferences for lower flow rates at high production rates, contradicting typical single-phase expectations.

economics at higher current densities where production rate gains significantly outpace energy efficiency losses. The Pareto analysis using production rate and energy productivity therefore balances the trade-offs between high selectivity at low current density and high throughput at elevated current density, providing a framework that accounts for both kinetic effects on selectivity and practical requirements for industrial operation.

The Pareto fronts for different organic fractions (Fig. 6C) show similar performance across all organic fraction ranges, but with important distinctions based on operating conditions. Operating above the solubility limit (> 0.09) is essential for achieving the highest production rates, but an organic fraction below the solubility limit is preferred for maximizing energy productivity at low current densities. This preference at low current densities likely reflects that excess AN promotes undesired side reactions such as tricyanohexane (TCH) formation when electrochemical demand is low, providing no mass transport benefit while increasing selectivity losses. As conditions shift toward high production rates (i.e., high current densities), operating above the solubility limit becomes beneficial due to increased mass transport limitations. Optimal conditions for high production rates occur at organic fractions corresponding to 1–2 times the solubility limit, with higher concentrations providing additional benefits but with diminishing returns. Notably, while the maximum production rate point occurs at organic fraction 0.3, a nearly equivalent production rate exists at organic fraction < 0.18 , suggesting that moderate organic fractions (1–2 times the solubility limit) may be preferred for industrial operation due to reduced reactant consumption, minimized side product formation, and simplified downstream separations.

The interelectrode gap has a significant effect on the Pareto fronts, as demonstrated in Fig. 6D. The 0.25 mm gap shows the worst performance across the Pareto front despite the advantage of reduced ohmic resistance from a smaller gap size. The decreased performance results from multiple detrimental effects: organic droplets and gas bubbles spanning the interelectrode gap create high-resistance zones and locally elevated current densities, while the reduced reactor volume limits bubble-induced convection that is critical for maintaining high performance at elevated current densities. The 1.0 mm gap performs substantially better than the 0.25 mm gap, particularly at low production rates and current densities, but shows diminishing advantages at high production rates. This performance pattern suggests a tradeoff between enhanced bubble-induced mixing from the larger volume and increased ohmic losses from the greater electrode separation. At low current densities, the mixing benefits clearly outweigh the ohmic penalties, but at high current densities and production rates, the increased ohmic losses balance out the convection benefits of the larger gap, causing performance to converge with the smaller gaps. However, the 0.5 mm gap demonstrates optimal performance, achieving the highest energy productivity at high production rates and superior performance across the entire Pareto front. This gap represents an optimal balance between minimizing ohmic resistance (favoring smaller gaps) and avoiding the negative effects of droplet and bubble spanning (favoring larger gaps). The 0.5 mm gap provides sufficient reactor volume for effective bubble-induced convection while maintaining low enough ohmic resistance to preserve energy efficiency, making it the preferred choice to maximize both throughput and energy performance.

Superficial velocity effects (Fig. 6E) reveal that lower velocities are more advantageous in multiphase flows, especially at high production rates. This result contradicts typical expectations where higher flow rates lead to enhanced mass transport and improved performance through forced convection. The increased performance for low superficial velocities can be attributed to the dominance of bubble-induced convection over forced convection effects in multiphase systems [13,31–33]. Lower velocities provide longer residence times that allow more time for bubble-induced mixing to enhance mass transport, an effect that outweighs any decrease in forced convection. Additionally, low superficial velocities tend to maintain droplet and slug flow regimes

that are conducive to high performance. This finding simplifies reactor operation by minimizing the need for high-energy pumping systems typically required in conventional electrochemical processes.

5. Conclusions

This systematic investigation of multiphase electrochemical systems using acrylonitrile electrohydrodimerization provides quantitative insights into the factors governing electrochemical performance in transport-limited organic synthesis. Through comprehensive flow visualization, data-driven modeling, and multi-objective optimization, our results suggest that bubble-induced convection—rather than inlet flow regime control—likely serves as the primary factor governing electrochemical performance in multiphase systems. Our flow regime characterization using Weber number maps established four distinct initial flow regimes (droplet, slug, parallel, and transition flows) across different operating conditions. However, these baseline flow patterns are disrupted during electrochemical operation due to gas bubble generation from electrode reactions. While flow regime effects are measurable, with droplet-like flows showing 10–15 % better selectivity than parallel flows, the magnitude of this effect is small compared to the influence of bubble-induced convection, interelectrode gap, and organic fraction, which maintain high selectivity at elevated current density. Importantly, the modest preference for droplet flows is advantageous from a practical standpoint, as droplet regimes are readily achievable through simple inline mixing. The dominance of bubble-induced convection enables multiphase systems to maintain high ADN selectivity (73–76 %) even at elevated current densities up to 300 mA cm^{-2} . This performance is likely to stem from bubble-induced mixing, which scales with electrochemical activity, generating enhanced mass transport when and where it is most needed. This creates a beneficial coupling where increased current density generates more bubbles, which in turn provide the mixing required to sustain high selectivity at those same elevated current densities.

Reactor geometry optimization reveals that the interelectrode gap represents a critical design parameter that must be carefully tuned to harness the full benefits of bubble-induced convection. Optimal performance is achieved at an intermediate gap of 0.5 mm, which achieves a balance between minimizing ohmic resistance (favoring smaller gaps) and avoiding negative effects from droplet and bubble spanning while providing sufficient volume for effective bubble-induced mixing (favoring larger gaps). Counterintuitively, the smallest gap (0.25 mm) does not provide the expected improved performance despite lower ohmic resistance, as the confined geometry limits bubble-induced convection and creates high-resistance zones when droplets and bubbles span the gap. The optimal operating conditions for maximum production rates—AN above solubility limits, 0.5 mm interelectrode gap, and low flow velocities—differ from conventional electrochemical reactor design principles. While higher flow velocities typically enhance forced convection in single-phase reactors, our multiphase reactor performed better at lower velocities where bubble-induced convection dominates, and longer residence times allow for more effective mixing. The multi-objective Pareto analysis provides practical insights for industrial implementation by revealing optimal tradeoffs between production rate and energy productivity, with maximum production rates of $5 \times 10^{-4} \text{ kg cm}^{-2} \text{ h}^{-1}$ achieved at 290 mA cm^{-2} , and maximum energy productivity of 0.4 kg kWh^{-1} at moderate current densities.

These results provide engineering design guidelines for multiphase electrochemical reactors that prioritize optimization of reactor geometry and current density to maximize bubble-induced convection, along with sufficient organic (above solubility limit) in the dispersed phase. The demonstrated ability to maintain high selectivity at elevated current densities, combined with simplified operational requirements, establishes multiphase operation as a viable strategy for overcoming mass transport limitations in organic electrosynthesis. These findings can inform the design of scaled electrochemical processes and encourage

broader exploration of multiphase systems for sustainable chemical manufacturing applications where conventional single-phase approaches face transport limitations.

CRedit authorship contribution statement

Casey K. Bloomquist: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Ziqi Zhang:** Investigation. **Eray S. Aydil:** Writing – review & editing, Supervision. **Miguel A. Modestino:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used Claude by Anthropic for editing assistance. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Miguel Modestino reports financial support was provided by National Science Foundation. Miguel A. Modestino is a co-founder and has a financial interest in Sunthetics, Inc., a start-up company in the machine learning optimization space. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We thank NYU Tandon School of Engineering and the National Science Foundation for their generous financial support. This material is based upon work supported by the National Science Foundation under grant no. 1943972.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ces.2025.172125>.

Data availability

Data will be made available on request.

References

- [1] D.S. Mallapragada, Y. Dvorkin, M.A. Modestino, D.V. Esposito, W.A. Smith, B.-M. Hodge, M.P. Harold, V.M. Donnelly, A. Nuz, C. Bloomquist, K. Baker, L. C. Grabow, Y. Yan, N.N. Rajput, R.L. Hartman, E.J. Biddinger, E.S. Aydil, A. D. Taylor, Decarbonization of the chemical industry through electrification: barriers and opportunities, *Joule* 7 (1) (2023) 23–41, <https://doi.org/10.1016/j.joule.2022.12.008>.
- [2] G.G. Botte, Electrochemical manufacturing in the chemical industry, *Interface Mag.* 23 (3) (2014) 49–55, <https://doi.org/10.1149/2.F04143if>.
- [3] P. De Luna, C. Hahn, D. Higgins, S.A. Jaffer, T.F. Jaramillo, E.H. Sargent, What would it take for renewably powered Electrosynthesis to displace petrochemical processes? *Science* 364 (6438) (2019) eaav3506 <https://doi.org/10.1126/science.aav3506>.
- [4] Z.J. Schiffer, K. Manthiram, Electrification and Decarbonization of the chemical industry, *Joule* 1 (1) (2017) 10–14, <https://doi.org/10.1016/j.joule.2017.07.008>.
- [5] N. Tanbouza, T. Ollevier, K. Lam, Bridging lab and industry with flow electrochemistry, *iScience* 23 (11) (2020) 101720, <https://doi.org/10.1016/j.isci.2020.101720>.
- [6] D.S.P. Cardoso, B. Šljukić, D.M.F. Santos, C.A.C. Sequeira, Organic Electrosynthesis: from laboratorial practice to industrial applications, *Org. Process. Res. Dev.* 21 (9) (2017) 1213–1226, <https://doi.org/10.1021/acs.oprd.7b00004>.
- [7] M. Yan, Y. Kawamata, P.S. Baran, Synthetic organic electrochemical methods since 2000: on the verge of a renaissance, *Chem. Rev.* 117 (21) (2017) 13230–13319, <https://doi.org/10.1021/acs.chemrev.7b00397>.
- [8] D.E. Blanco, M.A. Modestino, Organic Electrosynthesis for sustainable chemical manufacturing, *Trends Chem.* 1 (1) (2019) 8–10, <https://doi.org/10.1016/j.trechm.2019.01.001>.
- [9] E.J. Biddinger, M.A. Modestino, Electro-organic syntheses for green chemical manufacturing, *Electrochem. Soc. Interface* 29 (3) (2020) 43–47, <https://doi.org/10.1149/2.F06203IF>.
- [10] R. Mathison, A.L. Ramos Figueroa, C. Bloomquist, M.A. Modestino, Electrochemical manufacturing routes for organic chemical commodities, *Annu. Rev. Chem. Biomol. Eng.* 14 (1) (2023) 85–108, <https://doi.org/10.1146/annurev-chembioeng-101121-090840>.
- [11] Z.J. Oliver, D.J. Abrams, L. Cardinale, C.-J. Chen, G.L. Beutner, S. Caille, B. Cohen, L. Deng, M. Diwan, M.O. Frederick, K. Harper, J.M. Hawkins, D. Lehnher, C. Lucky, A. Meyer, S. Noh, D. Nunez, K. Quasdorf, J. Teli, S.S. Stahl, M. Schreiber, Scaling organic Electrosynthesis: the crucial interplay between mechanism and mass transport, *ACS Cent. Sci.* 11 (4) (2025) 528–538, <https://doi.org/10.1021/acscentsci.4c01733>.
- [12] T. Noël, Y. Cao, G. Laudadio, The fundamentals behind the use of flow reactors in electrochemistry, *Acc. Chem. Res.* 52 (10) (2019) 2858–2869, <https://doi.org/10.1021/acs.accounts.9b00412>.
- [13] C.K. Bloomquist, M. Dogan, J.S. Harris, B.D. Herzog, W.J. Tenn Iii, E.S. Aydil, M. A. Modestino, Understanding the effects of forced and bubble-induced convection in transport-limited organic Electrosynthesis, *React. Chem. Eng.* (2024) 930–939, <https://doi.org/10.1039/D3RE00579H>.
- [14] M. Atobe, H. Tateno, Y. Matsumura, Applications of flow microreactors in Electrosynthetic processes, *Chem. Rev.* 118 (9) (2018) 4541–4572, <https://doi.org/10.1021/acs.chemrev.7b00353>.
- [15] F. Marken, J.D. Wadhawan, Multiphase methods in organic Electrosynthesis, *Acc. Chem. Res.* 52 (12) (2019) 3325–3338, <https://doi.org/10.1021/acs.accounts.9b00480>.
- [16] P. Lu, R.C. Alkire, Mass transfer in parallel plate Electrolyzers with two-phase liquid-liquid flow, *J. Electrochem. Soc.* 131 (5) (1984) 1059–1067, <https://doi.org/10.1149/1.2115750>.
- [17] M.N. Kashid, A. Gupta, A. Renken, L. Kiwi-Minsker, Numbering-up and mass transfer studies of liquid-liquid two-phase microstructured reactors, *Chem. Eng. J.* 158 (2) (2010) 233–240, <https://doi.org/10.1016/j.cej.2010.01.020>.
- [18] M.N. Kashid, A. Renken, L. Kiwi-Minsker, Influence of flow regime on mass transfer in different types of microchannels, *Ind. Eng. Chem. Res.* 50 (11) (2011) 6906–6914, <https://doi.org/10.1021/ie102200j>.
- [19] M. Darekar, K.K. Singh, S. Mukhopadhyay, K.T. Shenoy, Liquid-liquid two-phase flow patterns in Y-junction microchannels, *Ind. Eng. Chem. Res.* 56 (42) (2017) 12215–12226, <https://doi.org/10.1021/acs.iecr.7b03164>.
- [20] L. Yang, M.J. Nieves-Remacha, K.F. Jensen, Simulations and analysis of multiphase transport and reaction in segmented flow microreactors, *Chem. Eng. Sci.* 169 (2017) 106–116, <https://doi.org/10.1016/j.ces.2016.12.003>.
- [21] C. Yao, Y. Zhao, G. Chen, Multiphase processes with ionic liquids in microreactors: hydrodynamics, mass transfer and applications, *Chem. Eng. Sci.* 189 (2018) 340–359, <https://doi.org/10.1016/j.ces.2018.06.007>.
- [22] E.A. López-Guajardo, G.M. Garza-Cantú, A. Marques-Camarena, E. Ortiz-Nadal, K. D.P. Nigam, A. Montesinos-Castellanos, Liquid-liquid flow pattern visualization and mapping in a millimetric size coiled tube, *Ind. Eng. Chem. Res.* 58 (2) (2019) 1075–1086, <https://doi.org/10.1021/acs.iecr.8b05315>.
- [23] M. Al-Azzawi, F.S. Mjalli, A. Husain, M. Al-Dahhan, A review on the hydrodynamics of the liquid-liquid two-phase flow in the microchannels, *Ind. Eng. Chem. Res.* 60 (14) (2021) 5049–5075, <https://doi.org/10.1021/acs.iecr.0c05858>.
- [24] A. Farahani, A. Rahbar-Kelishami, H. Shayesteh, Microfluidic solvent extraction of Cd(II) in parallel flow pattern: optimization, ion exchange, and mass transfer study, *Sep. Purif. Technol.* 258 (2021) 118031, <https://doi.org/10.1016/j.seppur.2020.118031>.
- [25] C. Yao, Y. Zhao, H. Ma, Y. Liu, Q. Zhao, G. Chen, Two-phase flow and mass transfer in microchannels: a review from local mechanism to global models, *Chem. Eng. Sci.* 229 (2021) 116017, <https://doi.org/10.1016/j.ces.2020.116017>.
- [26] G. Fu, F. Chen, L. Ni, D. Wei, J. Jiang, Liquid-liquid two-phase flow and size distribution of droplets in milli-channels: effect of gravity, *Int. J. Multiphase Flow* 150 (2022) 104005, <https://doi.org/10.1016/j.ijmultiphaseflow.2022.104005>.
- [27] D.E. Danly, Development and commercialization of the Monsanto electrochemical Adiponitrile process, *J. Electrochem. Soc.* 131 (10) (1984) 435C–442C, <https://doi.org/10.1149/1.2115324>.
- [28] M.M. Baizer, Electrolytic reductive coupling: I. Acrylonitrile, *J. Electrochem. Soc.* 111 (2) (1964) 215, <https://doi.org/10.1149/1.2426086>.
- [29] M.M. Baizer, C.R. Campbell, R.H. Fariss, R. Johnson, Adiponitrile Process 3193480 (July 6, 1965).
- [30] D.E. Blanco, A.Z. Dookhith, M.A. Modestino, Enhancing selectivity and efficiency in the electrochemical synthesis of Adiponitrile, *React. Chem. Eng.* 4 (1) (2019) 8–16, <https://doi.org/10.1039/C8RE00262B>.

- [31] A. Shah, J. Jorne, Mass transfer under bubble-induced convection in a vertical electrochemical cell, *J. Electrochem. Soc.* 136 (1) (1989) 144–153, <https://doi.org/10.1149/1.2096575>.
- [32] O.N. Cavatorta, U. Böhm, Mass transfer in electrolytic cells with gas stirring, *J. Appl. Electrochem.* 17 (2) (1987) 340–346, <https://doi.org/10.1007/BF01023300>.
- [33] D.J. Economou, R.C. Alkire, Two-phase mass transfer in channel Electrolyzers with gas-liquid flow, *J. Electrochem. Soc.* 132 (3) (1985) 601–608, <https://doi.org/10.1149/1.2113914>.